21-701: Discrete Mathematics

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Chapter 1

Graphs and Colourings

Much of this course will be about graphs, and a surprising amount of what we want to know about a graph can be phrased as a question about colouring it. We begin with the vocabulary, and then settle the first such question completely: which graphs admit a colouring with only two colours. We then ask the same question of hypergraphs, where it becomes considerably harder—even pinning down how many edges a 3-uniform hypergraph needs before two colours are no longer enough takes real work.

Notation.

- $[k]$ will denote $\{1, \ldots, k\}$.
- $\sim$ will denote the adjacency relation on graph vertices.

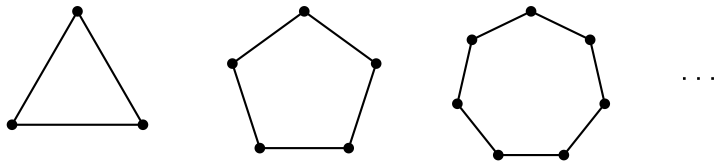
1.1 Colourings

Definition 1.1.1 (Colouring). Given a graph $G = (V, E)$, a k-colouring of G is a map $c : V \to [k]$ such that $u \sim v \implies c(u) \neq c(v)$.

1.1.1 2-Colourings of Graphs

Let's start with a motivating question: which graphs can be 2-coloured?

Any odd cycle graph can't be 2-coloured. That is, any graph that looks like



Let's study when odd cycles exist.

Lemma 1.1.2. *If G has an odd closed walk, then G has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbour, return).

Proof. Suppose G has an odd closed walk. Let W be such a walk of minimal length. Write $W = (x_1, \dots, x_i, \dots, x_j, \dots, x_\ell)$, with $x_\ell = x_1$. Suppose W is not a cycle. Then there must be some x_i, x_j such that $x_i = x_j$ even though $i \neq j$, ie, there must be some repetition. Then, $(x_1, \dots, x_i, x_{j+1}, \dots, x_\ell)$ and $(x_i, x_{i+1}, \dots, x_j)$ are both shorter walks than W. Moreover, one of them must be of odd length. This contradicts minimality of the length of W. So W must be a cycle. □

Here's a cool result.

Theorem 1.1.3. *A graph $G = (V, E)$ can be 2-coloured if and only if G has no odd cycle.*

Proof.

($\implies$) We've already seen that odd cycles can't be 2-coloured. So if G can be 2-coloured, it can't have an odd cycle. We've essentially seen this from the examples above.

($\impliedby$) We may assume G is connected, since otherwise we can colour each component separately. Fix some $v_0 \in V$. Define $c : V \to \{1, 2\}$ by

$$c(u) = \begin{cases} 2 & \text{if } \operatorname{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \operatorname{dist}(v_0, u) \text{ is even} \end{cases}$$

where $\text{dist}(\cdot, \cdot)$ denotes the shortest number of edges from one vertex to another.

We claim that this c is a valid 2-colouring.

Indeed, suppose it is not. Then, there must be some $u, v \in V$ such that $u \sim v$ and $c(u) = c(v)$. Now let's consider the two possible cases on the colour.

Suppose $c(u) = c(v) = 1$ for adjacent u, v. Then, there is a path of even length between v and v_0 and another path of even length between u and v_0. Concatenating the path from v to v_0 with the one from v_0 to u, and closing up with the edge uv, we get a closed walk of length even + even +1. This is odd, so by Theorem 1.1.2, G has an odd cycle.

The case $c(u) = c(v) = 2$ is identical: the two paths are now of odd length, so the closed walk they form with the edge uv has length odd + odd +1, which is odd again.

Either way, G has an odd cycle, contrary to hypothesis. So c must be a valid 2-colouring.

$\square$

1.1.2 Colourings of Hypergraphs

Colouring is not really a question about graphs. All the definition above asks of G is that we know which pairs of vertices may not share a colour, and nothing stops us forbidding larger sets instead.

Definition 1.1.4 (Hypergraph). Given a vertex set V, a **hypergraph** is a collection H of subsets of V. The sets in H are called **hyperedges**.

Definition 1.1.5 (Uniformity). A hypergraph H is k-**uniform** if $|f| = k$ for all $f \in H$.

The following is thus obvious.

Proposition 1.1.6. *A graph is a 2-uniform hypergraph.*

Definition 1.1.7. A proper colouring of a hypergraph H is an assignment $c : V \to [r]$ such that $\forall f \in H$, $|c(f)| \geq 2$.

We say "no edge is monochromatic".

We can ask ourselves the following question: **"Which 3-uniform hypergraphs are 2-colourable?**

For each k, define $m(k)$ to be the smallest m such that $\exists$ some k-uniform hypergraph with m edges that is not 2-colourable. We can show that $m(2) = 3$. But what is $m(3)$? That is, how many 3-edges do I need to prevent 2-colourability?

Let H be a 3-uniform hypergraph that's not 2-colourable. Assume, further, that H has at most 6 edges (each being a subset of size 3).

First, we'll show that we can reduce, without loss of generality, to the case where H has exactly 6 vertices. Indeed, if H has vertices x, y that are not in any common edge, then I can define H' by "merging" x and y. So H' has one fewer vertex than H. This H' cannot be 2-colourable: if it were, then that colouring would extend to H by using the colour of the merged vertex at both x and y, which would make H 2-colourable. And when H has more than 6 vertices, such a pair always exists.

An edge has 3 vertices, so it covers the $\binom{3}{2} = 3$ pairs among them, and H has at most 6 edges, so at most

$$6 \cdot \binom{3}{2} = 18$$

pairs are covered in total. But if H has $n \geq 7$ vertices, there are $\binom{n}{2} \geq \binom{7}{2} = 21$ pairs to cover. So some pair is left uncovered, and merging it brings us one vertex closer to 6.

So now assume that H has exactly 6 edges and 6 vertices and is not 2-colourable.

Bibliography

These lecture notes are based heavily on the following references:

[1] Nicolas Bourbaki. *Groupes et Algèbres de Lie. Chapitre 1*. 1st ed. Éléments de Mathématique. Édition originale publiée par Hermann, Paris, 1972. Springer Berlin, Heidelberg, 2007. ISBN: 978-3-540-35337-9. DOI: https://doi.org/10.1007/978-3-540-35337-9.

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